Tactile FLORA

Fingertip force as text for a vision-language reward model

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A vision-only reward model cannot see what the fingers feel

FLORA-Rewarder (NVIDIA, Qwen3.5-27B) watches a video mosaic and scores task success 1 to 5. It has no contact channel.

- **Grasp or no grasp.** An empty gripper closes, lifts and moves exactly like a loaded one.
- **Hold or touch.** A brief contact looks like a grasp on camera; the force says it ended early.
- **Weak grip, re-grasp, slip.** Visually unremarkable, obvious in the force trace.
- **Small views.** At 96 px per camera a cube is a few pixels wide; the lift is invisible.

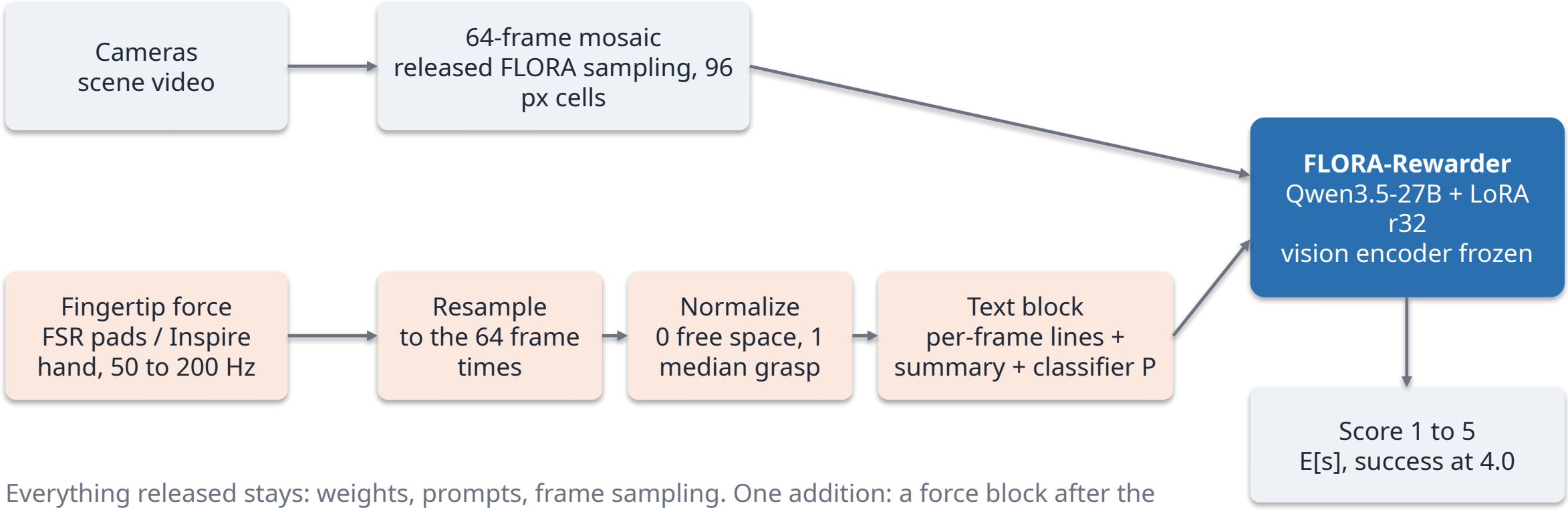
Released FLORA on a Baxter success

“the cube remained on the table in the final frames”

The cube was 20 cm in the air, held for 2.9 s.

Rationale from the released model on rollout ep0000 (writeup/baxter-rollouts/README.md).

The idea: render force as text, put it in the prompt, fine-tune with LoRA



Everything released stays: weights, prompts, frame sampling. One addition: a force block after the video, introduced by one sentence that names its columns. Matched control: same training with the block omitted.

The force block the model reads

```
Force (normalized: 0 free space, 1 median grasp, max 3):
summary R: contacts=1 longest=6.00s total=6.00s peak=1.83 hold=5.67s
   holdF=1.55 late/early=0.70 low=0% force-only classifier
P(success)=0.76
t=0.00 L none | pad R F=0.00 D=0.00 dF=0.00 g=0.00 c=0
t=0.25 L none | pad R F=0.00 D=0.00 dF=0.00 g=0.00 c=0
...
t=12.50 L none | pad R F=1.62 D=0.21 dF=0.03 g=0.71 c=1
```

HRDexDB training clip inspire_dftp/cactus2/0, abbreviated. Per-trial tare, two most loaded pads, sign-corrected closure.

Why text: no new encoder, no change to the released model; the resampler keeps the raw slice so an encoder can be added later.

- **F** total grip force; D pad asymmetry; dF its rate
- **g** hand closure, 0 open to 1 closed; c contact flag (F above 5% of grasp)
- **summary** label-free: contact segments, longest and total contact, peak, hold time and force, late/early hold ratio, low-force fraction
- **P(success)** a 120-tree, depth-2 gradient-boosted classifier over 31 force statistics, fitted on the 350 training clips; training clips see out-of-fold values
- **budget** at most 3,000 tokens per clip (largest seen: 2,700)

Every claim is checked against two controls and two diagnostics

1

Matched no-force fine-tune

Same base, same 350 clips, same seed, epochs, learning rate, batch and LoRA; block omitted. A gain over it is a force gain, not a fine-tuning gain.

2

Shuffled force

The tactile model with each clip's block replaced by another clip's. A gain that survives shuffling is not a force gain.

3

Force removed

The tactile adapter with no block at all: does it still rank by what it learned from video?

4

Video blanked

Frames mid-gray, force only: what the force carries on its own.

Held-out sets are object-disjoint from training, split by object hash before any model ran, and scored once with fixed settings (threshold 4.0, no checkpoint selection).

Four stages: each one fixed what the previous one showed was wrong

1

Pilot

32 / 16 HRDexDB clips

No force benefit. Audit: tiny sample, unaligned sensor and video clocks.

2

Expanded v2

618 clips, aligned clocks, frozen object-disjoint splits 350 / 136 / 115

Still no benefit: 82.4% vs 81.6%. Block found nearly uninformative.

3

Bugs found

Inverted closure sign, force saturated at 37.5 units, contact flag always on

Classifier on rendered lines: AUROC 0.57; on raw tared force: 0.78.

4

v3 (primary)

Tared, peak-ranked, sign-corrected, summary line, classifier P

85.3% / AUROC 0.926 on dev; 72.2% / 0.800 on test.

5

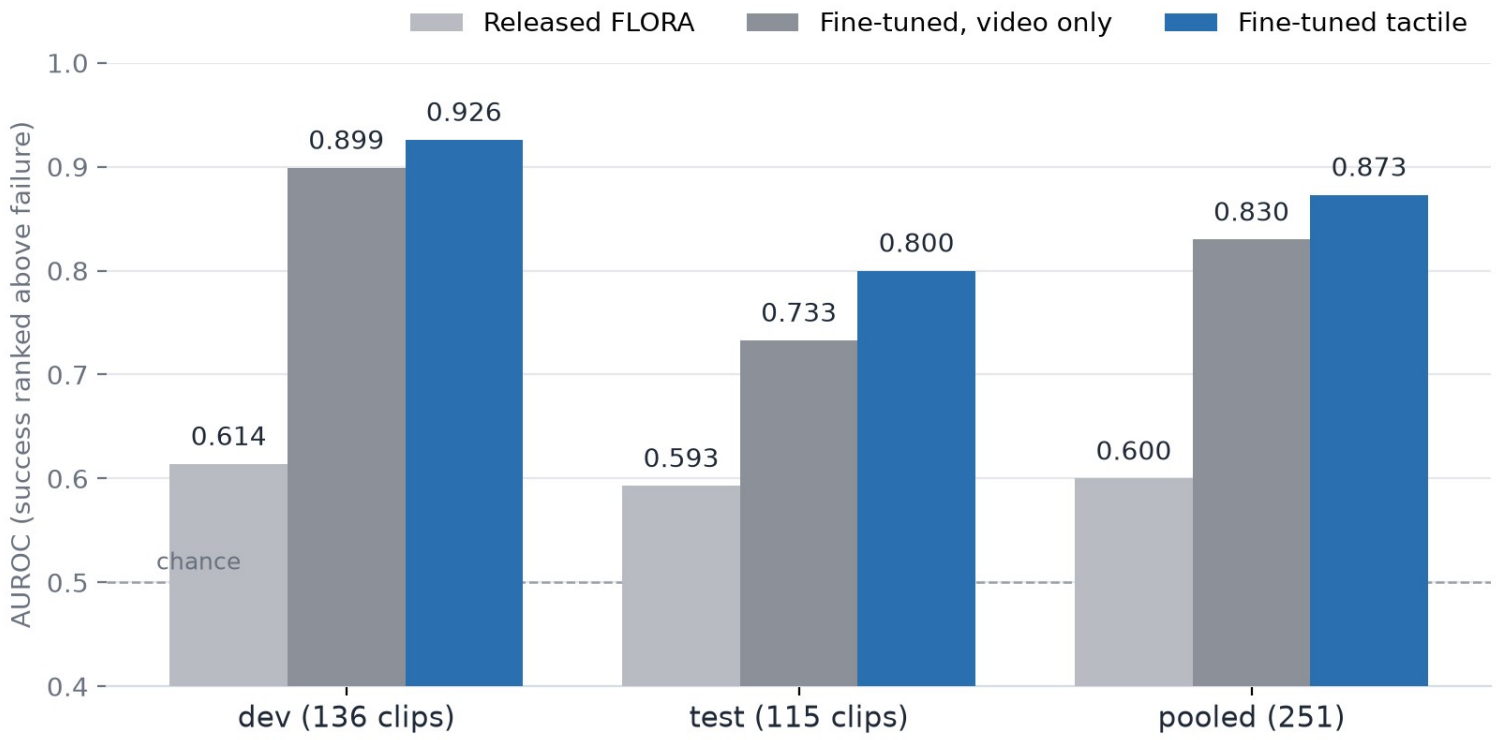
Baxter

FSR fingertips, 50 demos, ACT policy, 10 rollouts

Zero-shot: force takes AUROC from 0.31 to 1.0.

Training: released FLORA base, four epochs, seed 17, LoRA r32/a64, lr 2e-5, batch 16, 88 updates, two AMD MI210s. Only the force representation changed between v2 and v3.

HRDexDB: the tactile fine-tune ranks best on every held-out split



Accuracy at the 4.0 threshold

Model	dev	test	pooled
Released	66.2%	67.0%	66.5%
Video only	82.4%	69.6%	76.5%
Tactile	85.3%	72.2%	79.3%

MAE on dev (mean |score − truth| on the 1 to 5 scale, times 100): 150 released, 71 video only, 53 tactile.

251 clips over 32 objects never seen in training. Pooled: +12.8 accuracy points and +0.27 AUROC over released FLORA; +2.8 points and +0.043 AUROC over the matched video-only fine-tune.

The ranking gain over the video-only fine-tune is significant; the accuracy gain is not

+0.043

AUROC, tactile minus video-only, pooled over 251 clips

+0.001 to +0.080

object-level bootstrap 95% interval, p about 0.02

26 vs 19

clips fixed vs clips broken relative to the video-only model

Per split

- dev: +0.026 AUROC (not significant); 12 wins, 8 losses
- test: +0.067 AUROC (p about 0.04); 14 wins, 11 losses
- accuracy +2.8 points overall, exact McNemar p = 0.37

Against released FLORA

- +12.8 accuracy points and +0.27 AUROC pooled
- dev MAE 150 to 53 (mean absolute error on the 1 to 5 scale, times 100)
- 251 clips cannot certify a 3-point accuracy gain; about 1,000 would

The model reads the force: shuffle it and the gain disappears

Tactile adapter on dev (136)	Accuracy	AUROC	MAE ×100
Real force	85.3%	0.926	53
Shuffled force (another clip's block)	73.5%	0.824	109
Video blanked (force only)	53.7%	0.775	121
Video-only fine-tune, for reference	82.4%	0.899	71

25 : 9

paired clips lost : won when the force is shuffled, p = 0.009

- **Shuffling costs 12 points.** A gain that survived shuffling would not be a force gain; this one does not survive it.
- **Force alone is weak.** With the video blanked the model still ranks above chance (0.775) but calls almost everything a failure.
- **Both channels are used.** Neither video nor force alone reaches 0.926; the fine-tuned model combines them.

The force settles the clips the video cannot call, and leaves the rest alone

Clips, split by the video-only model's confidence	Video only	Tactile	Gain
Uncertain: E[s] within 0.75 of the threshold (101 clips)	66.3%	**72.3%**	+6.0
Confident: the other 150 clips	83.3%	**84.0%**	+0.7
dev only, uncertain band (46 clips)	71.7%	**82.6%**	+10.9

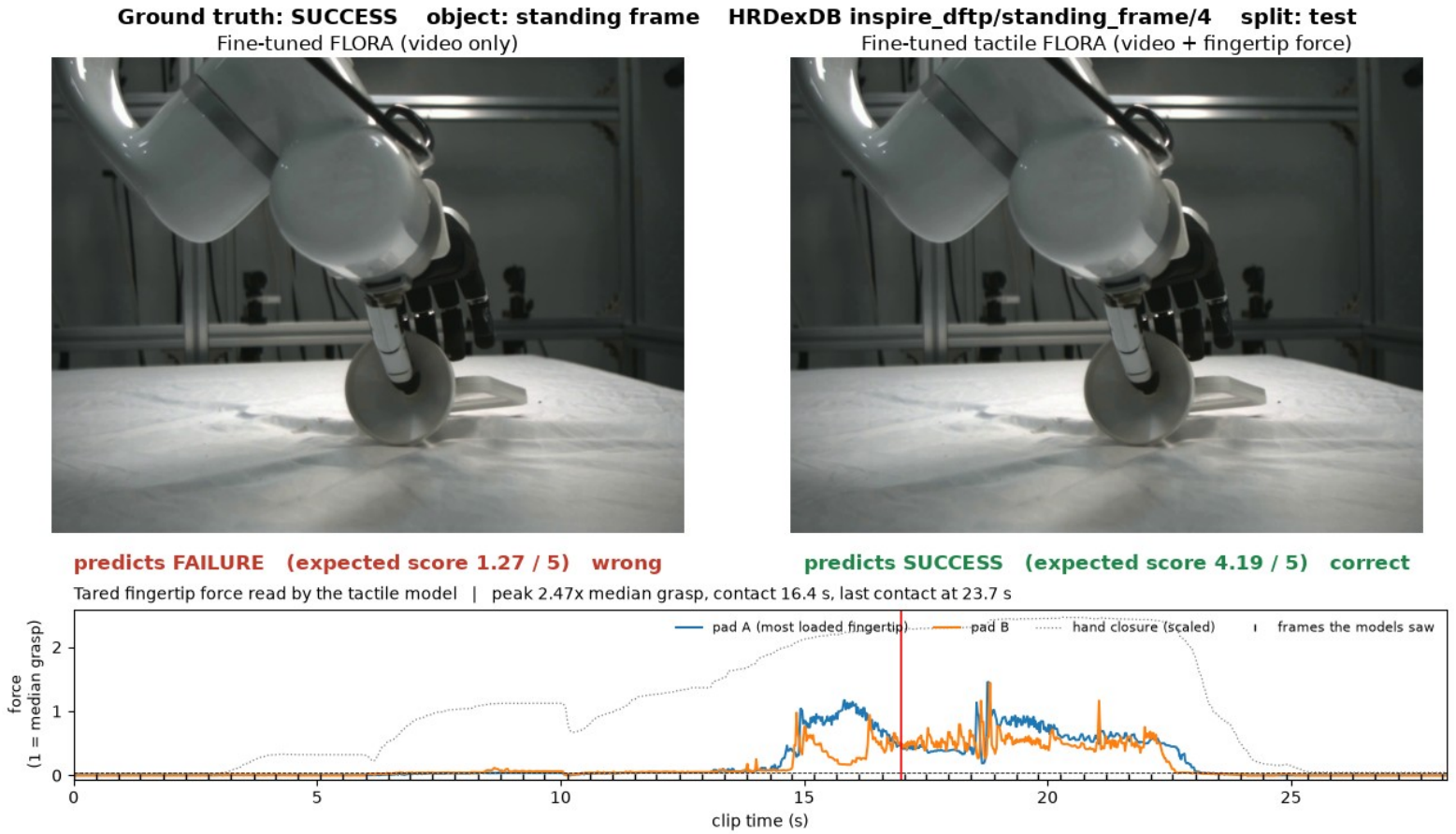
What the wins look like

The video-only model under-calls successes whose hold is visually unremarkable (long, steady contact above the median force) and over-calls failures where the hand closes convincingly but the force shows a weak grip, a re-grasp, or contact that ends early.

Stacking check

A force-only classifier alone scores 76.5% / 0.777 on test, level with both VLMs on accuracy; a logistic stack of video score, tactile score and classifier reaches 88.2% / 0.928 on dev. The VLM, not the sensor, is the bottleneck.

A win: standing frame #4, a success the camera scored 1.27



HRDexDB test clip

1.27

video only: failure

4.19

tactile: success (truth)

Force: peak 2.5x the median grasp, 16.4 s of contact, last contact at 23.7 s of 28.2 s.

All 26 such clips are rendered as side-by-side videos in writeup/hrdexdb/videos/; the 19 the video-only model wins are listed in writeup/hrdexdb/README.md.

Baxter: our own fingertips, our own policy, zero-shot evaluation

Sensor

Four thin-film FSR pads in a 3.3 V divider, Teensy 4.0 at 200 Hz, 12-bit ADC, clock-fitted into ROS. Left gripper only; raw counts, resting offset 92.

Demos

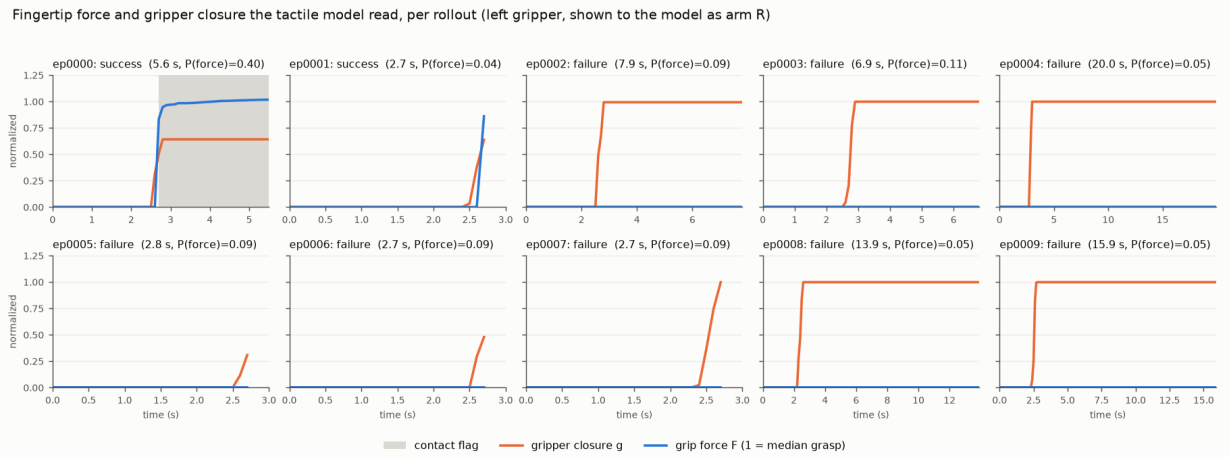
50 hand-guided left-arm cube picks, 15 fps head and wrist cameras at 640 x 360, no tactile. Cut at the first sign of the gripper re-opening.

Policy

ACT (LeRobot 0.6.1): chunk 30, 15 action steps, ResNet-18, VAE; 50k steps on five RTX 2080 Ti. Rollouts use the 30k checkpoint (held-out L1 0.028).

Rollouts

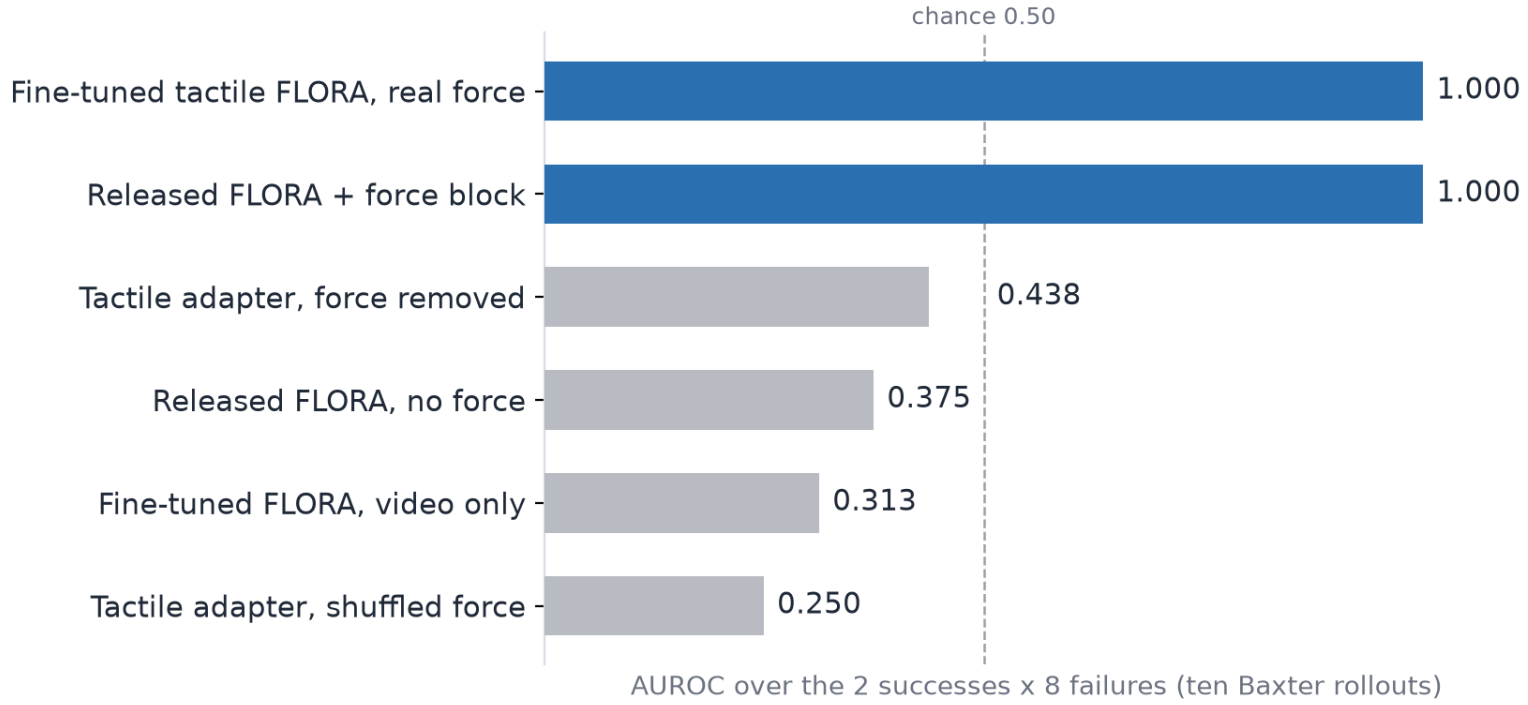
Ten live runs at 30 Hz with joint-jump guard, fingertip box and speed cap, tactile recorded. Operator labels: ep0000 and ep0001 succeeded, eight failed.



Force and closure the tactile model read, per rollout

Only ep0000 shows a clean grasp signature: closure stops at 0.64 on the cube and force holds at the grasp level for 2.9 s. ep0001 has one contact frame at the very end. The eight failures closed on nothing (closure 1.0, force 0), and five of them then lift the empty gripper exactly as a success would.

Baxter: the force block takes the ranking from below chance to perfect



0.31 → 1.00

AUROC, fine-tuned FLORA without and with the force block

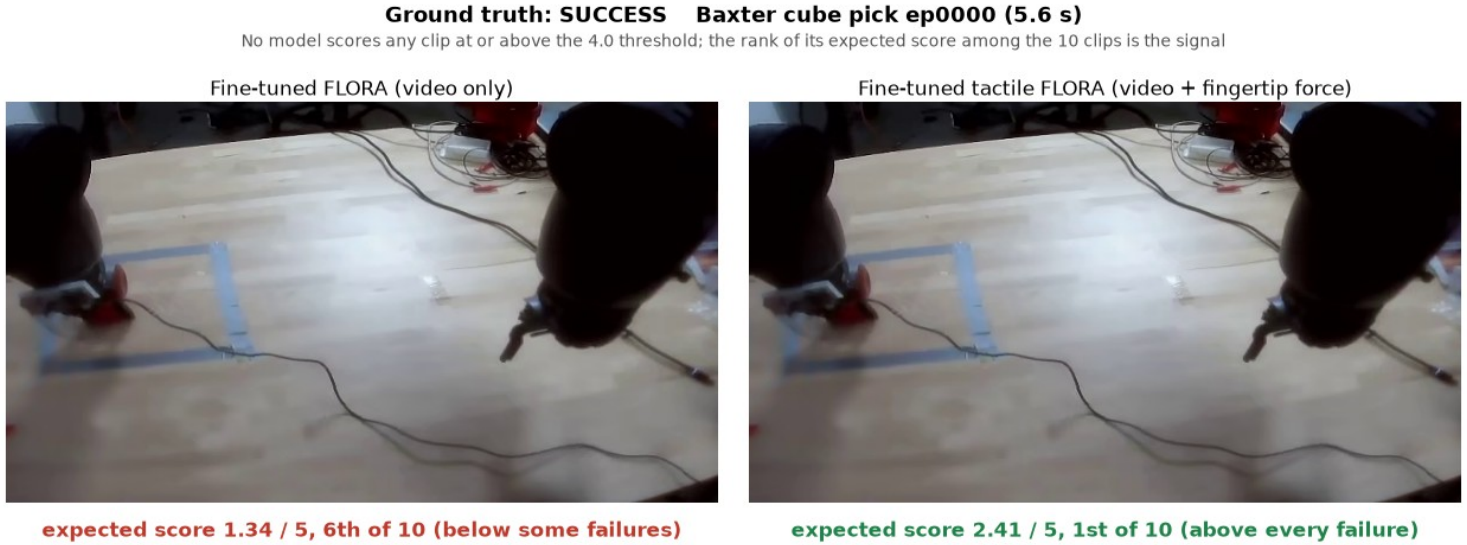
0.375 → 1.00

released FLORA without and with the block, no training at all

Every condition has accuracy 0.80, the majority class: no model scores a success at or above 4.0, so the signal is the ranking. Shuffled force 0.25, force removed 0.44, force-only classifier 0.50.

Ten clips, two successes: 16 ordered pairs. A direction, not a significance result. Slurm job 6490, two MI210s, 8 min.

ep0000: the video models cannot see the lift, the force block can



Episode	Video only	Tactile
ep0000 success	1.34, 6th of 10	**2.41, 1st**
ep0001 success	1.05, 8th	**1.60, 2nd**
ep0003 empty lift	1.99, 1st	**1.09, 3rd**
ep0008 empty lift	1.75, 2nd	**1.04, 6th**

Expected score and rank among the ten clips. The tactile model's lowest success (1.60) sits 0.51 above its highest failure (1.09).

ep0003 and ep0008 are the video-only model's two top scores: the arm closes, lifts and moves exactly as in a success. With its force block removed the tactile adapter also puts them on top (2.97, 2.31).

Side-by-side videos for the seven clips where the tactile model wins by at least 0.1: writeup/baxter-rollouts/videos/.

Caveats

- **Sample size.** 251 held-out clips cannot certify a 3-point accuracy gain; the AUROC gain is the significant one. Baxter: ten clips, two positives, no significance claim.
- **Control provenance.** The video-only comparison is the v2 control adapter; the v3 control retrain was preempted. Inputs are identical, so only training noise is unmeasured.
- **A small classifier rides in the block.** Its probability is one line the VLM reads; the raw force lines stay. It cannot explain Baxter (its own AUROC there is 0.5).
- **Undertrained.** Loss floors by update 20 of 88: the target is a template plus one digit. Both fine-tunes are data-limited.
- **96 px views.** All video models saw tiny thumbnails; higher resolution would help every one of them and may shrink the force gain.
- **Baxter specifics.** Normalization fitted on the ten test clips (label-free); ep0001 rests on one line of the block; uncalibrated FSR counts; left pads shown as the right-hand pads the adapter knew.

Tactile FLORA outperforms regular FLORA at judging grasps

vs released FLORA

+12.8 pts, +0.27 AUROC on HRDexDB; 0.375 to 1.0 on Baxter

vs video-only fine-tune

+2.8 pts, +0.043 AUROC (p about 0.02); +6 pts where video is unsure; 0.31 to 1.0 on Baxter

the force is causal

shuffled: 25 lost / 9 won (p = 0.009), Baxter 0.25; removed: 0.44

What's next

- More held-out clips (about 1,000) to certify the accuracy gain, and a second training seed
- Higher-resolution video input, then re-measure how much force still adds
- Calibrated fingertip sensors on both Baxter grippers and more rollouts, including induced slips
- More embodiments: the same text block works for any force stream